[INCH–POUND]

A–A–59789/21A
21 October 2016
SUPERSEDING
A–A–59789/21
21 October 2011

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 5 PIECE, FOR COLD PLATE APPLICATIONS, .225 X .260 INCH BODY SIZE, WEDGE MOUNTING BODY, WITH TWO OR NO MOUNTING HOLES, SCREW ACTUATED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59789.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes herein).



Example: AA59789/21WR55S is the PIN for a clear chemical film finished, 5.3 inch (134.6 mm) long card holder. The card holder also features two tapped mounting holes for use with 2–56 UNC 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in its installed position and prevent loosening or movement as a result of shock and vibration. It shall also provide a thermal transfer path from the circuit card assembly to the cold plate.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A–A–59789.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59789.

Actuating screw hex drive socket. The dimension for hex drive socket shall be .094 inch (2.38 mm) across flats for mounting options "N", "S", "R", "P", and "T".

Cold plate slot width. The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59789).

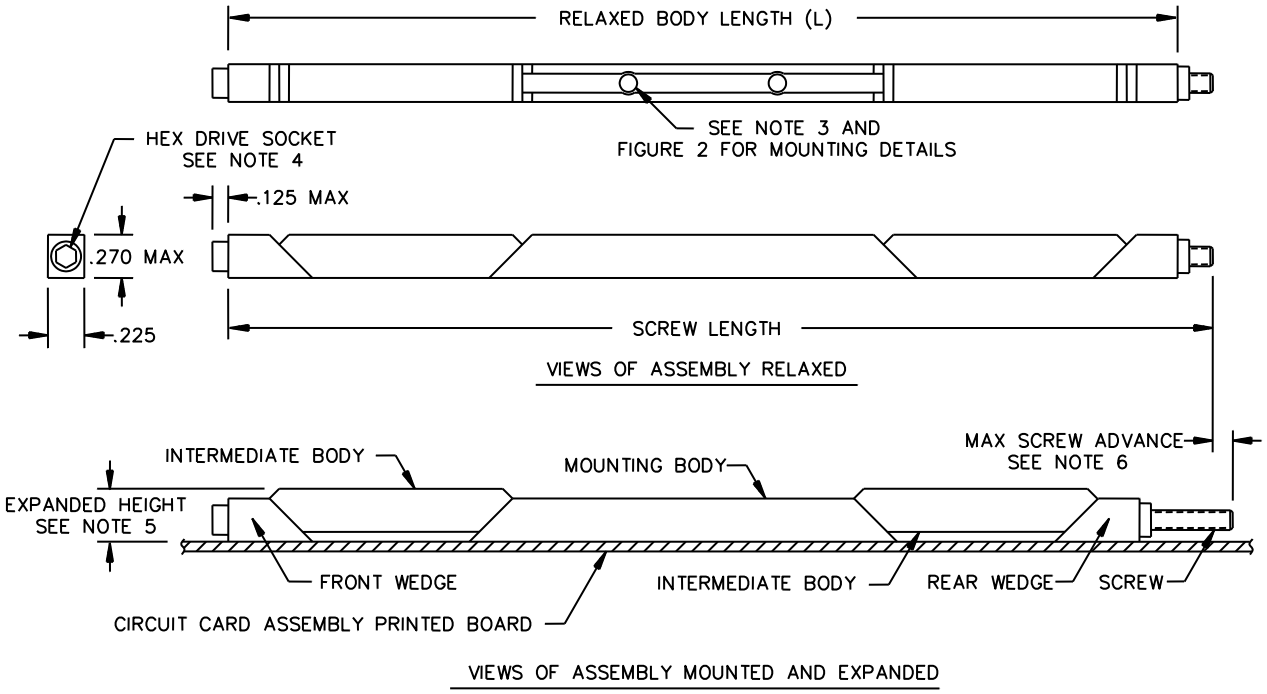
Installation torque. The recommended nominal installation torque is as follows: 6 inch-pounds (68 N-cm) for assemblies of configuration "A", "C", "F", or "W" and 7.2 inch-pounds (81 N-cm) for assemblies of configurations "D", "J", "L", or "N".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

Configuration. The configuration of a card holder shall be as specified in table I. The details of a particular configuration consist of those on figures 1 and 2, and may include those on figure 3 or 4.

TABLE I. Configuration.

Configuration	Mounting body style (see figure 2)	Mounting options	Applicable figures	Hardware options
A	A	P, S, R, T	1 and 2	No added hardware options
C	C	N, S, R, T		
F	A	P, S, R, T	1, 2, and 3	Lockwasher and flat washer
W	C	N, S, R, T		
N	A	P, S, R, T	1, 2, and 4	Screw self-lock element
L	C	N, S, R, T		
J	A	P, S, R, T	1, 2, 3, and 4	Lockwasher, flat washer, and screw self-lock element
D	C	N, S, R, T		



Inches	mm	Inches	mm	Inches	Mm
.125	3.18	.225	5.72	.30	7.6
.140	3.56	.270	6.86	.900	22.86

FIGURE 1. Relaxed and expanded dimensions (two mounting hole version shown).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Screw and alignment spring not shown. See table II and figure 2 for hole spacing requirements.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm).
5. Mounting body style "A" is capable of safely expanding to .330 inch (8.38 mm) and mounting body style "C" is capable of expanding to .325 inch (8.26 mm) minimum.
6. Maximum screw advance is .12 inch (3.0 mm) for mounting body style "A" and .14 inch (3.6 mm) for mounting body style "C".

FIGURE 1. Relaxed and expanded dimensions – Continued.

Finish. The finish designator shall be as specified in A–A–59789. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "E" (electroless nickel), "H" (hard black anodize), or "R" (clear chemical film).

Length, expanded and relaxed dimensions. The length designator shall be as specified in A–A–59789. The lengths available for this CID specification sheet are listed in table II. The length, expanded and relaxed dimensions shall be as specified on figure 1.

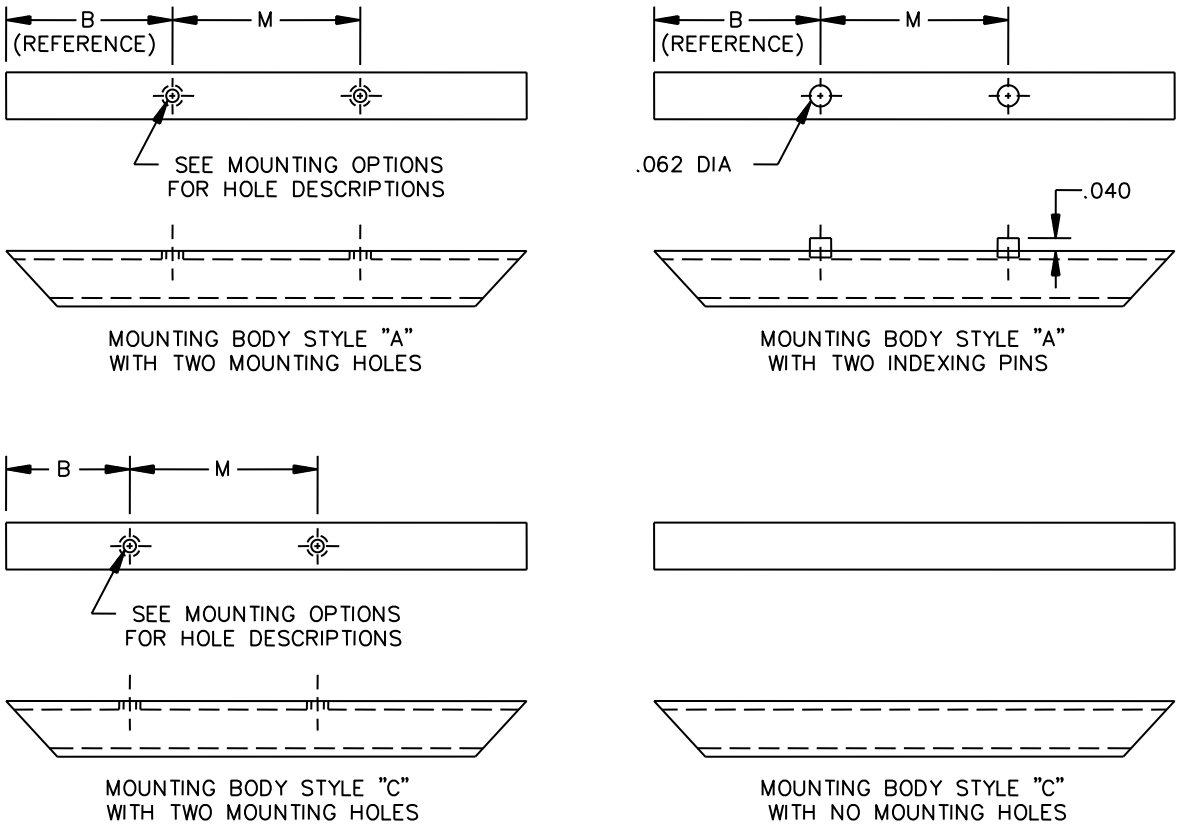
TABLE II. Assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "L" ±.02 (0.5)	Mounting body style	Dimension "A" ±.01 (0.3)	Dimension "B" ±.02 (0.5)	Dimension "M" 2/
50	4.8 (121.9)	A	1.50 (38.1)	.30 (7.6)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
55	5.3 (134.6)	A	2.00 (50.8)	.55 (14.0)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
60	5.8 (147.3)	A	2.50 (63.5)	.80 (20.3)	.900
		C	1.60 (40.6)	.35 (8.9)	.900
65	6.3 (160.0)	A	3.00 (71.2)	1.05 (26.7)	.900
		C	1.60 (40.6)	.35 (8.9)	.900

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for general information only.
2/ Tolerance for the hole spacing is ±.005 inch (0.51 mm).

Mounting body style (see figure 2 and table II). Two mounting body styles are available on this CID specification sheet, style A or style C. The primary difference in the two mounting body styles are the location of mounting holes on the mounting body as depicted on figure 2. Mounting body style A always has two locations for fasteners or indexing pins centered on the mounting body that are .900 inch (22.86 mm) apart. Mounting body style C can have two holes or no holes on the mounting body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.



Inches	mm
.040	1.02
.062	1.57

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. See table II for dimensions B and M.

FIGURE 2. Mounting body style details.

Mounting. The mounting designators shall be as specified in A–A–59789. The mounting options available for this CID specification sheet are as follows: "N" (no mounting holes; mounting body style C only), "P" (Ø.062 x .040 index pins; mounting body style A only), "R" (rivet mount holes with counterbore and countersink), "S" (tapped 2–56 holes), or "T" (tapped 0–80 holes). See figure 1 for mounting hole spacing requirements. Card holders using mounting option "P" and "R" may be shipped unassembled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

Rivet mount holes. The holes used for rivet mounting shall be .066/.073 inch (1.68/1.85 mm) diameter, countersunk 100 degrees by .140 inch (3.56 mm) diameter through holes with a counterbore of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

Rivets. This card holder uses rivet style A as specified in A–A–59789 when rivet mounting is used.

Hardware options. Card holders can have the following hardware options: No added hardware options, lock washer and flat washer (see figure 3), screw self locking element (see figure 4), or a combination of both lock washer and flat washer and screw self locking element. See table I for the correct PIN configuration identifier.

No added hardware. The base configuration of the card holder is as depicted by figures 1 and 2. Card holders requiring no added hardware options shall include configuration identifier "A" or "C" in the PIN (see classification, table I and notes section herein).

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include configuration identifier "F" or "W" in the PIN (see classification, table I and notes section herein).

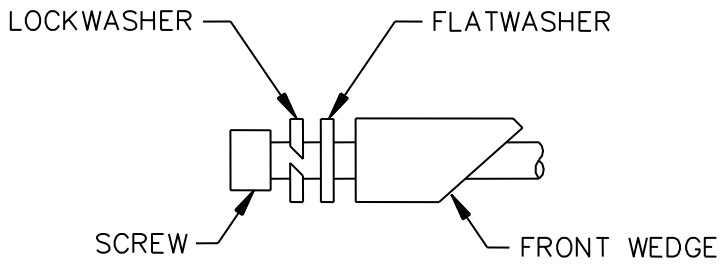


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for increased resistance to loosening of the card holder assembly from shock vibration. The screw self-locking element shall be in accordance with A–A–59789. Card holders requiring a screw self-locking element shall include configuration identifier "L" or "N" in the PIN (see classification, table I and notes section herein).

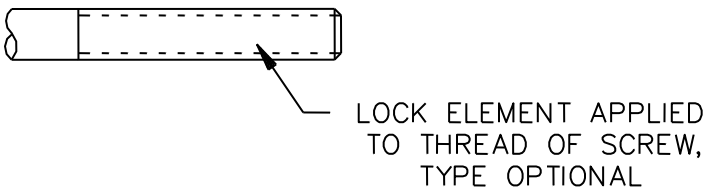


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include configuration identifier "D" or "J" in the PIN (see classification, table I and notes section herein).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59789 – Holder, Electrical Card, Wedge Retainers, 5 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

Ordering data. Ordering data shall be as specified in A–A–59789.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
5BG68	American Circuit Card Retainers, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@accrmfg.com URL: http//www.accrmfg.com
61081	Pentair Technical Products (formerly Birtcher) 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 E-mail: schroff.us@pentair.com URL: http://schroff.pentair.com

Part number supersession data. These CID specification sheet PINs supersede the following manufacturer's part numbers in table III and on DESC Drawing 89064 (table IV) as shown. The CID PINs listed in table III are only for length designator "50". See tables V and VI for CID PIN construction using other available lengths for this CID specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

TABLE III. Commercial part number supersession data.

PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50S	40–5–10–B	CB50S	G5225BA–4.80T2
AB50R	40–5–10–B–R	CB50R	G5225BA–4.80H
AB50T	40–5–10–B–T	CB50T	G5225BA–4.80T0
AB50P	40–5–10–B–P	CB50N	G5225BA–4.80
FB50S	40–5–10–B–LF	WB50S	G5225WBA–4.80T2
FB50R	40–5–10–B–LF–R	WB50R	G5225WBA–4.80H
FB50T	40–5–10–B–LF–T	WB50T	G5225WBA–4.80T0
FB50P	40–5–10–B–LF–P	WB50N	G5225WBA–4.80
NB50S	40–5–10–B–L	LB50S	G5225BA–4.80T2L
NB50R	40–5–10–B–R–L	LB50R	G5225BA–4.80HL
NB50T	40–5–10–B–T–L	LB50T	G5225BA–4.80T0L
NB50P	40–5–10–B–P–L	LB50N	G5225BA–4.80L
JB50S	40–5–10–B–LF–L	DB50S	G5225WBA–4.80T2L
JB50R	40–5–10–B–LFRL	DB50R	G5225WBA–4.80HL
JB50T	40–5–10–B–LFTL	DB50T	G5225WBA–4.80T0L
JB50P	40–5–10–B–LFPL	DB50N	G5225WBA–4.80L
AC50S	40–5–10	CC50S	G5225CG–4.80T2
AC50R	40–5–10–R	CC50R	G5225CG–4.80H
AC50T	40–5–10–T	CC50T	G5225CG–4.80T0
AC50P	40–5–10–P	CC50N	G5225CG–4.80
FC50S	40–5–10–LF	WC50S	G5225WCG–4.80T2
FC50R	40–5–10–LF–R	WC50R	G5225WCG–4.80H
FC50T	40–5–10–LF–T	WC50T	G5225WCG–4.80T0
FC50P	40–5–10–LF–P	WC50N	G5225WCG–4.80
NC50S	40–5–10–L	LC50S	G5225CG–4.80T2L
NC50R	40–5–10–R–L	LC50R	G5225CG–4.80HL
NC50T	40–5–10–T–L	LC50T	G5225CG–4.80T0L
NC50P	40–5–10–P–L	LC50N	G5225CG–4.80L
JC50S	40–5–10–LF–L	DC50S	G5225WCG–4.80T2L
JC50R	40–5–10–LF–R–L	DC50R	G5225WCG–4.80HL
JC50T	40–5–10–LF–T–L	DC50T	G5225WCG–4.80T0L
JC50P	40–5–10–LF–P–L	DC50N	G5225WCG–4.80L

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AP50S	40–5–10–N	CE50S	G5225EN–4.80T2
AP50R	40–5–10–N–R	CE50R	G5225EN–4.80H
AP50T	40–5–10–N–T	CE50T	G5225EN–4.80T0
AP50P	40–5–10–N–P	CE50N	G5225EN–4.80
FP50S	40–5–10–N–LF	WE50S	G5225WEN–4.80T2
FP50R	40–5–10–N–LF–R	WE50R	G5225WEN–4.80H
FP50T	40–5–10–N–LF–T	WE50T	G5225WEN–4.80T0
FP50P	40–5–10–N–LF–P	WE50N	G5225WEN–4.80
NP50S	40–5–10–N–L	LE50S	G5225EN–4.80T2L
NP50R	40–5–10–N–R–L	LE50R	G5225EN–4.80HL
NP50T	40–5–10–N–T–L	LE50T	G5225EN–4.80T0L
NP50P	40–5–10–N–P–L	LE50N	G5225EN–4.80L
JP50S	40–5–10–N–LF–L	DE50S	G5225WEN–4.80T2L
JP50R	40–5–10–N–LFRL	DE50R	G5225WEN–4.80HL
JP50T	40–5–10–N–LFTL	DE50T	G5225WEN–4.80T0L
JP50P	40–5–10–N–LFPL	DE50N	G5225WEN–4.80L
AH50S	40–5–10–B3	CH50S	G5225BH–4.80T2
AH50R	40–5–10–B3–R	CH50R	G5225BH–4.80H
AH50T	40–5–10–B3–T	CH50T	G5225BH–4.80T0
AH50P	40–5–10–B3–P	CH50N	G5225BH–4.80
FH50S	40–5–10–B3–LF	WH50S	G5225WBH–4.80T2
FH50R	40–5–10–B3–LFR	WH50R	G5225WBH–4.80H
FH50T	40–5–10–B3–LFT	WH50T	G5225WBH–4.80T0
FH50P	40–5–10–B3–LFP	WH50N	G5225WBH–4.80
NH50S	40–5–10–B3–L	LH50S	G5225BH–4.80T2L
NH50R	40–5–10–B3–R–L	LH50R	G5225BH–4.80HL
NH50T	40–5–10–B3–T–L	LH50T	G5225BH–4.80T0L
NH50P	40–5–10–B3–P–L	LH50N	G5225BH–4.80L
JH50S	40–5–10–B3–LFL	DH50S	G5225WBH–4.80T2L
JH50R	40–5–10–B3LFRL	DH50R	G5225WBH–4.80HL
JH50T	40–5–10–B3LFTL	DH50T	G5225WBH–4.80T0L
JH50P	40–5–10–B3LFPL	DH50N	G5225WBH–4.80L

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AR50S	40–5–10–CC	CR50S	G5225CC–4.80T2
AR50R	40–5–10–CC–R	CR50R	G5225CC–4.80H
AR50T	40–5–10–CC–T	CR50T	G5225CC–4.80T0
AR50P	40–5–10–CC–P	CR50N	G5225CC–4.80
FR50S	40–5–10–CC–LF	WR50S	G5225WCC–4.80T2
FR50R	40–5–10–CC–LFR	WR50R	G5225WCC–4.80H
FR50T	40–5–10–CC–LFT	WR50T	G5225WCC–4.80T0
FR50P	40–5–10–CC–LFP	WR50N	G5225WCC–4.80
NR50S	40–5–10–CC–L	LR50S	G5225CC–4.80T2L
NR50R	40–5–10–CC–R–L	LR50R	G5225CC–4.80HL
NR50T	40–5–10–CC–T–L	LR50T	G5225CC–4.80T0L
NR50P	40–5–10–CC–P–L	LR50N	G5225CC–4.80L
JR50S	40–5–10–CCLF–L	DR50S	G5225WCC–4.80T2L
JR50R	40–5–10–CCLFRL	DR50R	G5225WCC–4.80HL
JR50T	40–5–10–CCLFTL	DR50T	G5225WCC–4.80T0L
JR50P	40–5–10–CCLFPL	DR50N	G5225WCC–4.80L

1/ The CID PINs listed are only for length designator "50".
2/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
3/ Mounting body style A.
4/ Mounting body style C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

TABLE IV. Supersession information.

Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DESC Drawing 89064	Superseding (new) CID PIN
89064–50AN	AA59789/21AC50N	89064–50ANW	AA59789/21FC50N
89064–50AR	AA59789/21AC50R	89064–50ARW	AA59789/21FC50R
89064–50AS	AA59789/21AC50S	89064–50ASW	AA59789/21FC50S
89064–50AT	AA59789/21AC50T	89064–50ATW	AA59789/21FC50T
89064–60AN	AA59789/21AC60N	89064–60ANW	AA59789/21FC60N
89064–60AR	AA59789/21AC60R	89064–60ARW	AA59789/21FC60R
89064–60AS	AA59789/21AC60S	89064–60ASW	AA59789/21FC60S
89064–60AT	AA59789/21AC60T	89064–60ATW	AA59789/21FC60T
89064–65AN	AA59789/21AC65N	89064–65ANW	AA59789/21FC65N
89064–65AR	AA59789/21AC65R	89064–65ARW	AA59789/21FC65R
89064–65AS	AA59789/21AC65S	89064–65ASW	AA59789/21FC65S
89064–65AT	AA59789/21AC65T	89064–65ATW	AA59789/21FC65T
89064–50EN	AA59789/21CC50N	89064–50ENW	AA59789/21WC50N
89064–50ER	AA59789/21CC50R	89064–50ERW	AA59789/21WC50R
89064–50ES	AA59789/21CC50S	89064–50ESW	AA59789/21WC50S
89064–50ET	AA59789/21CC50T	89064–50ETW	AA59789/21WC50T
89064–60EN	AA59789/21CC60N	89064–60ENW	AA59789/21WC60N
89064–60ER	AA59789/21CC60R	89064–60ERW	AA59789/21WC60R
89064–60ES	AA59789/21CC60S	89064–60ESW	AA59789/21WC60S
89064–60ET	AA59789/21CC60T	89064–60ETW	AA59789/21WC60T
89064–65EN	AA59789/21CC65N	89064–65ENW	AA59789/21WC65N
89064–65ER	AA59789/21CC65R	89064–65ERW	AA59789/21WC65R
89064–65ES	AA59789/21CC65S	89064–65ESW	AA59789/21WC65S
89064–65ET	AA59789/21CC65T	89064–65ETW	AA59789/21WC65T

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

PIN length examples. The CID PINs listed in table IV are for all available standard card holder lengths for this specification sheet. However, only one specific finish, mounting and configuration are listed (see PIN example for a break-down of the codes).

TABLE V. Examples of PINs with available length designators (two hole mounting bodies).

PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50S	40–5–10–B	CB50S	G5225BA–4.80T2
AB55S	40–5–11–B	CB55S	G5225BA–5.30T2
AB60S	40–5–12–B	CB60S	G5225BA–5.80T2
AB65S	40–5–13–B	CB65S	G5225BA–6.30T2

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
2/ Other lengths are available on request.
3/ Mounting body style A.
4/ Mounting body style C.

TABLE VI. Examples of PINs with available length dimensions (no mounting holes in bodies).

PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 3/ CAGE 61081	PIN designator AA59789/21	Vendor similar designator or type part number 1/ 2/ 4/ CAGE 5BG68
AB50P	40–5–10–B–P	CB50N	G5225BA–4.80
AB55P	40–5–11–B–P	CB55N	G5225BA–5.30
AB60P	40–5–12–B–P	CB60N	G5225BA–5.80
AB65P	40–5–13–B–P	CB65N	G5225BA–6.30

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59789.
2/ Other lengths are available on request.
3/ Mounting body style A with indexing pins in the mounting body.
4/ Mounting body style C with no holes in the mounting body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project 5998–2016–021

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:27Z
Check the source to verify that this is the current version before use.